

Optical detection of plastic waste through computer vision

Islomjon Shukhratov^a, Andrey Pimenov^a, Anton Stepanov^a, Nadezhda Mikhailova^a,
Anna Baldycheva^{b,*}, Andrey Somov^a

^a Skolkovo Institute of Science and Technology, Moscow, 121205, Russia

^b University of Exeter, Exeter, EX4 4QF, UK

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ABSTRACT

An incredibly wide range of plastic products and enormous volumes of generated plastic wastes makes sorting technologies highly important. In order to enhance the efficiency and accuracy of sorting process, this article proposes an Internet of Video Things solution based on the deep learning algorithms for image recognition of plastic waste on a moving conveyor belt and embedded intelligence. In particular, the state-of-the-art object detection models, including Faster R-CNN, RetinaNet and YOLOv8 are used. Target categories of plastic are Polyethylene Terephthalate (PET) and Polypropylene (PP). Furthermore, we implemented quantization techniques for trained models on a commercial off-the-shelf embedded system for fast processing time. We achieved a high mean Average Precision (mAP) metric of 77.74% and accuracy of 95.67% on a test set and are fine-tuned and optimized for the deployment on an Nano embedded system providing 20 frames per second. This research contributes to the field of application of Internet of Things by demonstrating the efficacy of deep learning algorithms run on the embedded system in the industrial plastic waste sorting process. The findings highlight the practical applicability of these algorithms and offer insights into resource management and recycling practices.

1. Introduction

Plastic waste has become a global environmental concern, with its widespread use and improper disposal leading to significant ecological and health consequences. Nearly 400 million tons of plastic are produced each year (Wamba et al., 2023). The content of plastics in municipal waste composition is constantly increasing (Ilyinykh et al., 2013). It is estimated that single used plastic products (bags, food utensils, packaging materials, etc.) contribute about 50 percent of that plastic waste (Duru et al., 2019). Regardless of waste collection system, this category is subject to sorting which means the physical removal from the heterogeneous mix of waste components. Effective management of plastic waste necessitates accurate sorting technologies (Ruj et al., 2015).

Manual sorting involves people categorizing and removing plastic waste based on physical characteristics such as shape, color, and labeling. This method is labor-intensive and prone to human errors. Therefore, it is challenging for handling the massive volume of generated plastic waste (Taneepanichskul et al., 2022).

The state-of-the-art solutions are energy-hungry and costly to install, operate and maintain. Current commercial technologies for automated sorting of waste are based mainly on Near Infrared (NIR) (Masoumi et al., 2012), X-ray fluorescence (XRF) (Gundupalli et al., 2017), and Visual (VIS) based solutions (Karaca et al., 2013, Lubongo & Alexandridis, 2022, Vogt et al., 2021, Araujo-Andrade et al., 2021). Commonly used VIS technologies are the prismcoupled color camera measuring colors (red, green, blue) and visible spectrometry (Lu & Chen, 2022).

The recent advancements in Internet of Things (IoT) open wide vista for industrial applications. In particular, there is a promising direction towards the Internet of Video Things (IoVT) (Nisha, 2023) aiming to enhance the opportunities of deep learning and computer vision (Lu & Chen, 2022, Sergiyenko et al., 2023, Hammoud et al., 2023) together with embedded intelligence (Amadeo et al., 2023). This paradigm offers promising solutions for automating the detection and classification of plastic waste (Lu & Chen, 2022). Nevertheless, due to the lack of an appropriate dataset used to waste detection in sorting facilities, creates a number of issues to develop robust and efficient deep learning

* Corresponding author.

E-mail addresses: Islomjon.Shukhratov@skoltech.ru (I. Shukhratov), andrey.pimenov@skoltech.ru (A. Pimenov), a.stepanov@skoltech.ru (A. Stepanov), N.Mikhailova@skoltech.ru (N. Mikhailova), a.baldycheva@exeter.ac.uk (A. Baldycheva), a.somov@skoltech.ru (A. Somov).

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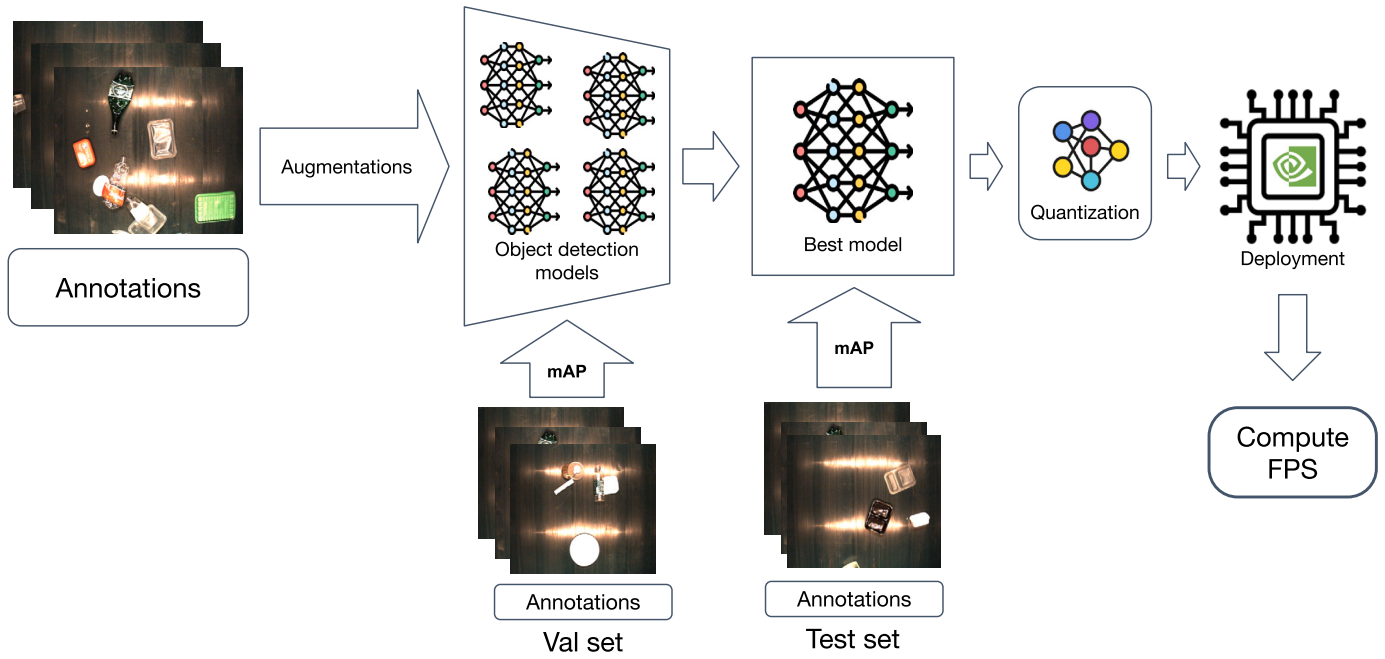


Fig. 1. Pipeline of proposed model.

methods to accurately sort plastic waste. Furthermore, inaccurate sorting critically reduces the quality of end products made from recycled polymers (Hopewell et al., 2009).

Contribution of this research is an accurate deep learning algorithm for the detection of plastic waste at a high-speed (3 m/s) conveyor belt. The proposed solution ensures the data collection of plastic waste, in particular Polyethylene Terephthalate (PET) and Polypropylene (PP), through video cameras with the help of custom made lightning system, use of deep learning models to detect it and further neural network compression for deploying it on an embedded system as depicted in Fig. 1. We make a comparative study of high-precision models and notable accuracy on diverse neural network architectures. This solution allows for the local data analysis without overloading the network and cloud infrastructure. In terms of advantages of the proposed solution, the use of an RGB camera with deep-learning algorithms achieving promising results without an expensive spectral camera.

The novelty of this research is twofold. (i) The data collection was organized in a similar to industry conditions. Thus, a conveyor system from the plastic sorting industry provides exactly the same speed as in the shop floor of the recycling factories equal to 3 m/s. A developed lighting system contributes to reducing the exposition time for a video camera and therefore increases its productivity. According to the literature review, other projects involving an embedded system rely on non-industrial datasets which makes practical feasibility of those solutions in industrial conditions unlikely. (ii) Spectral cameras are used in sorting industry to distinguish different types of plastics among each other. This research relies on a RGB-camera with the deep-learning algorithms achieving promising results without an expensive spectral camera. The proposed approach is characterized by economic reasoning. Two types of plastics (Polyethylene Terephthalate (PET) and Polypropylene (PP)) are investigated. However, the proposed solution can be scaled to other types of plastic.

For this research, we conducted a number of experiments and comparative analysis focusing on diverse neural network architectures like Faster R-CNN, RetinaNet and cutting-edge YOLOv8, all tailored for optimized waste detection. In addition, the efforts included the collection of a novel dataset that substantively contributes to the academic advancement of Solid Waste Management (SWM). Another important milestone of this research emerged as we compressed our deep learning models strategically boosting processing efficiency on an embedding system.

Typically, the object detection models in deep learning undergo rigorous evaluation, often measured by the robust metric of Mean Average Precision (mAP). The models excelled in this regard achieve an impressive 77.74% precision and an equally notable accuracy 95.67%. At the same time, on the embedded system Jetson Nano, our models exhibited remarkable efficiency, processing at a commendable rate of 20 frames per second. The efforts resulted in an improved framework with the potential to greatly enhance the field of waste detection efficiency.

This article is organized as follows. Section 2 explores the machine learning and deep learning methods used in waste classification, detection and segmentation. Section 3 outlines the methodology used to develop and evaluate a plastic waste detection system. Also, it covers the data collection, deep learning model selection and evaluation metrics. Section 4 presents the outcomes of the binary plastic waste detection task using the object detection models along with the deployment on the embedded system. In Section 5 we evaluate the achieved results and compare them with the state-of-the-art. Finally, Section 6 provides conclusions, limitations of this research and further insights to better handling the plastic waste sorting with deep learning.

2. Related works

2.1. Machine learning algorithms for waste recognition

Recent advances in deep learning can drastically change how plastic waste recognition is performed with cheaper devices and other sources of consumption, high accuracy and efficiency. Plastic waste sorting using deep learning algorithms belongs to the image-based sorting that relies on the identification of plastic material according to its shape, color, texture and size captured by a camera and processed with a particular computer vision algorithm. Image recognition consists of different tasks such as classification, localization, detection and segmentation.

Sakr et al. (2016) compared a Support Vector Machines (SVM) algorithm with the Convolutional neural networks where the SVM achieved high result. A dataset consisting of three categories, i.e. metal, paper and plastic, was trained in the SVM algorithm with a bag of features that finds the features using a window with size 8×8 by traversing over all training dataset; whilst AlexNet (Krizhevsky et al., 2012) was chosen to train the data with a neural network. Their results show that SVM considerably surpassed AlexNet with an average overall classes

95% vs. 83% of accuracy, respectively. The authors admit that the low results achieved by AlexNet were due to decreasing the size of images by 8 times of original size as well as overfitting issues that occurred during the training process. Alternatively, Yang and Thung (2016) compared SVM with Scale-Invariant Feature Transform (SIFT) features and AlexNet-like CNN. SIFT algorithms detect, express and match the local features, so-called keypoints in images (Lowe, 1999), where the object in a new image is detected by individually comparing its keypoints with the features in the existing dataset based on their Euclidean distance. The prediction of test images consisting of six classes (i.e. glass, paper, cardboard, plastic, metal and trash) showed better results in SVM with mean 58% precision, 60% recall and 63% accuracy compared to AlexNet with 18.7% precision, 19.3% recall and 22% accuracy. The weak performance of AlexNet was due to SVM being simpler than the CNN algorithm, as well as CNN being a data-hungry algorithm that requires a substantially large quantity of data points per class. Though these works show the considerable strength of SVMs over CNN, since then, many robust deep learning algorithms have been developed to tackle the waste sorting issues.

Aral et al. (2019) compared several deep vision networks to classify the TrashNet dataset (Yang & Thung, 2016), namely Xception (Chollet, 2017), MobileNet (Howard et al., 2017), DenseNet (Huang et al., 2017) and InceptionResnet (Szegedy et al., 2016) networks. After several times of fine-tuning, DenseNet121 and DenseNet169 networks gained the most significant accuracy score of 95%. The same approach was made with EfficientNet (Tan & Le, 2019) and ResNet (He et al., 2016) architectures by Masand et al. (2021) to classify TrashNet (Yang & Thung, 2016) and proposed ScrapNet dataset that contained additional plastic waste images. Classification of TrashNet dataset achieved 98% of accuracy, while ScrapNet only 93%. The high accuracy in TrashNet dataset was achieved because of choosing the decent hyperparameters, network architecture and training in a more extensive set of TrashNet that allowed algorithms to learn a large number of features.

A working prototype of plastic sorting using a deep learning algorithm was proposed by Bobulski and Kubanek (2021). Basically, a system consists of an RGB camera, where the individual waste is captured, a microcomputer with a computer vision preprocessing software and deep learning model, and air nozzles controlled by the output of AI software. They trained a 15-layer CNN architecture with the WaDaBa dataset (Bobulski & Piatkowski, 2018) and achieved 74% of accuracy due to the relatively weak model as well as compression of input images to 120×120 for quick computation during inference.

In terms of segmentation, Wang et al. (2020) proposed the segmentation of an object with intensity and depth information of an image. Two consecutive methods were applied: getting the initial segmentation of an object; and zoomed the prediction together with the conditional random field, depth and affinities are applied to create pixel-accurate segmentation. The proposed method was integrated into the state-of-the-art segmentation models like DeepLabv3 (Chen et al., 2017) and achieved 88% of Intersection over Union (IoU) metric in their proposed MJU dataset. One of the limitations of the work lies in the dataset itself: only the indoor photographs of individuals holding litter occurrences in their hands are included in the dataset which is far from the real-life plastic waste sorting.

Sousa et al. (2019) presented two methods of learning to detect plastic waste with their own dataset with 19 classes. The first method uses the traditional object detection algorithms like faster R-CNN to detect the objects and make classification from the output in a simple CNN classifier. Another method, which is called hierarchical learning, consists of merging several classes into a few “mother” classes and predicting only these classes while making further outputs as inputs to a special dedicated CNN classifier to get the final classes. The first method achieved 81.4% in mAP metrics, while the hierarchical learning achieved 86% in mAP. Though achievements in detection are reasonably good, in the real world scenarios it could take much time to make an inference in such models or large computation power.

Proença and Simões (2020) proposed a new dataset for litter detection “TACO” that contains mostly images of litter in the wild taken by smartphones. The authors used Mask R-CNN to detect and segment the objects in the image achieving 15.9 class average precision.

Majchrowska et al. (2022) used EfficientNet-B2 and EfficientDet-B2 network models to classify and detect litter objects in the dataset that combines several images from TACO, MJU and other datasets (Lynch, 2018, Yang & Thung, 2016).

In addition to scholarly works related to plastic waste sorting using computer vision, there are also industrial solutions like TOMRA, Pellenc and Steiner that are already utilizing the deep learning methodologies. One of them, AMPRobotics (2023), developed by AMP Neuron which claims to use computer vision to recognize plastic waste in different positions and forms: stacked, folded or damaged, the recognized object later is removed by a robotic hand. Terex (2023) is yet another company that focuses on the waste sorting domain. They, similarly to AMP robotics, use the computer vision algorithms to detect and a robotic arm and/or picker to remove the objects from the conveyor belt. Moreover, they report that they can sort objects that include plastic, scrap metals, wood, mixed inert and stones. To sum up, there are huge advances in deep learning techniques that are being applied to plastic waste sorting.

2.2. Datasets

Yang and Thung (2016) presented the TrashNet, the manually collected dataset in 2017 for trash classification. The dataset contains a total of 2400 RGB images for 6 categories: glass, plastic, paper, cardboard, metal and trash. Each category has 400-500 images with approximate 400 px width. In order to ensure diversity within the dataset, the authors employed a variety of pose and lighting levels. To further enhance the dataset’s efficacy they used the data augmentation techniques, such as random rotation of images, adjustments in brightness, and both scaling and shearing transformations. By employing these techniques they aimed to create a more robust and reliable dataset for object classification. Another relevant experimental research on a dataset collection for unmanned vending machines is reported in Zhang et al. (2023).

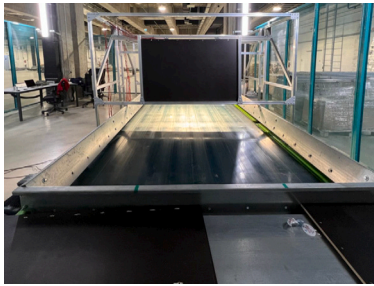
Bobulski and Piatkowski (2018) presented the WaDaBa Dataset for waste classification. The dataset encompasses 4000 RGB images, pertaining to six distinct types of plastic, alongside other classes that consist of objects displaying various levels of deformation, dirtiness, as well as clean and underformed objects. To ensure that the dataset captured the wide spectrum of real-world scenarios, the authors utilized scattered light from a fluorescent lamp and warm LED light reflector. This methodology ensured that the obtained dataset had a diverse range of lighting conditions, and was therefore better equipped to detect and classify objects in varied settings.

The Kaggle repository became a significant resource for open-source datasets in recent years and provides a number of open datasets for waste classification. The Garbage Classification Data (Vignesh, 2022) contains 2467 images for six distinct classes of waste, namely cardboard, glass, metal, paper, plastic, and trash. The images have an average width of 400 px. Another dataset, the Waste Classification data (Sekar, 2022), comprises 22,500 images scraped from the web and classifies waste into two categories: organic and recyclable. These images have an average width of 250 px. Later, an updated version of this dataset was released with an additional 2900 images, introducing a third class of waste: non-recyclable. The Drinking Waste Classification dataset (Serezhkin, 2019) features 9640 images and encompasses four distinct classes of waste, including aluminum cans, glass bottles, PET and HDPE bottles. The images in this dataset have an average width of 550 px and partially use the TrashNet dataset to increase the dataset’s robustness.

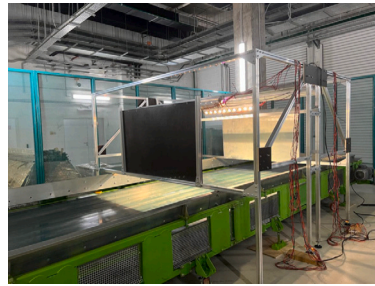
Sousa et al. (2019) introduced the Labeled Waste in the Wild dataset which includes 1002 photographs of food trays taken by smartphones, for detecting bounding boxes for 19 classes with an average width of 3500 px. Another dataset MJU-Waste v1.0 (Wang et al., 2020) was pre-

Table 1
Comparative study of plastic waste datasets.

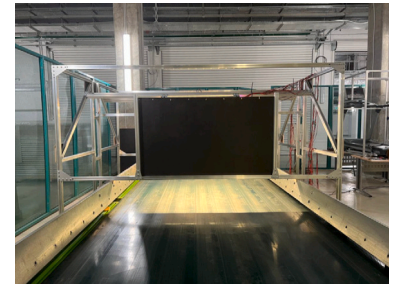
Dataset	# images	Classes	Task	Width	Year
TrashNet	2400	glass, paper, cardboard, plastic, metal, trash	Classification	400px	2017
WaDaBa	4000	6 types of plastics and other classes	Classification	400px	2018
Garbage Classification Data	2467	same as TrashNet	Classification	400px	2017
Waste Classification Data	22,500	2: organic and recyclable	Classification	250px	2018
OpenLitterMap	100,000+	100+	Classification	N/A	2019
VN-trash Dataset	5904	organic, inorganic and medical	Classification	N/A	2019
Labeled Waste in the Wild	1002	19	Detection	3500px	2019
Drinking Waste Classification	9640	4: aluminium, PET, glass, PET, HDPE bottles	Detection	550px	2019
Waste Classification Data v2	22,500+	3: organic, recyclable, non-recyclable	Classification	300px	2019
TACO	1500	28	Detection	2500px	2020
MJU-Waste v1.0	2475	1	Segmentation	500px	2020
TrashCan v1	7212	22	Instance Segmentation	600px	2020
Proposed dataset	1900	2: PET and PP	Detection	600px	2023



(a)



(b)



(c)

Fig. 2. Industrial conveyor machine and aluminum platform with lighting. (a) - front view, (b) - side view and (c) - back view.

sented by Wang et al. which contains 2475 images for segmentation of a single class - waste. Images were captured by taking a picture of a waste with an average width of 500 px, sometimes while being held by people. Additionally, Hong et al. (2020) presented the TrashCan v1 dataset consisting of 7212 images with 22 classes for image instance segmentation use case. However, since the dataset mainly addresses marine pollution, the images mostly contain the trash that is captured in water. The summary of available waste datasets is presented in Table 1.

The reviewed datasets have undoubtedly been helpful in the development of robust and reliable machine learning algorithms for waste detection. However, they are characterized by some limitations. One significant drawback is the lack of similarity to real-world conveyor scenarios where the waste typically arrives in batches. In such settings, the effectiveness of the trained algorithms can be reduced, resulting in suboptimal performance.

To overcome these challenges and improve the accuracy of waste detection models, a novel dataset has been collected within this research, which will be discussed in the next section. This dataset aims at addressing the shortcomings of existing datasets by providing more representative and diverse samples that closely mimic the conveyor belt's real-world scenarios. This approach can lead to a more accurate and reliable models, reducing the chances of false positives and false negatives.

3. Methods

3.1. Data collection

In order to combat the challenges that previous datasets encounter, it was decided to collect a novel dataset that will represent the real-life scenario of a conveyor belt with plastic waste. In this section a detailed description of the collected dataset is provided.

An industrial conveyor machine for plastic waste sorting is installed in the laboratory, the part of which is surrounded by aluminium platform where industrial camera and lighting are installed vertically to conveyor belt to capture waste that moves along the belt. The velocity of the belt ranges from 0.5 m/s to 3 m/s, the belt has a 2 m width and a 6 m length. The camera is installed in the middle of the conveyor and captures images of 1.2 m width and 1 m height. The experimental setup is shown in Fig. 2.

The camera has 1.3 MP. It is placed at a height of 1-1.4 m above the belt and captures images continuously in RGB color space, which are then saved in JPG format with a resolution of 1280×1024 pixels. On both sides of camera, lighting is installed to give more brightness and to ensure reasonable reflection two panels painted in white color are installed opposite of each light source. However, since the speed of conveyor belt is high, the camera's exposure time is set to 900 ms which may lead to some images being duplicated or without a waste sample on it.

To collect the dataset for waste detection, two classes of plastic waste were utilized, namely PET and PP. PET mainly consists of bottles while PP has varying types. However, the availability of plastic waste samples was limited. Also, there were some repetitive samples. To overcome this issue and increase the variability of the samples, we divided the dataset into the training and testing samples. This approach helped in ensuring that the dataset included a diverse set of samples for both the training and the testing purposes. In doing so, the deep learning algorithms trained on this dataset can be more robust and generalize better to unseen data. Furthermore, the use of two classes of plastic waste provided a good representation of the most commonly found plastic materials in the waste stream.

The choice of using two classes can be explained by the factor that collecting a large number of datasets with all plastic classes along with

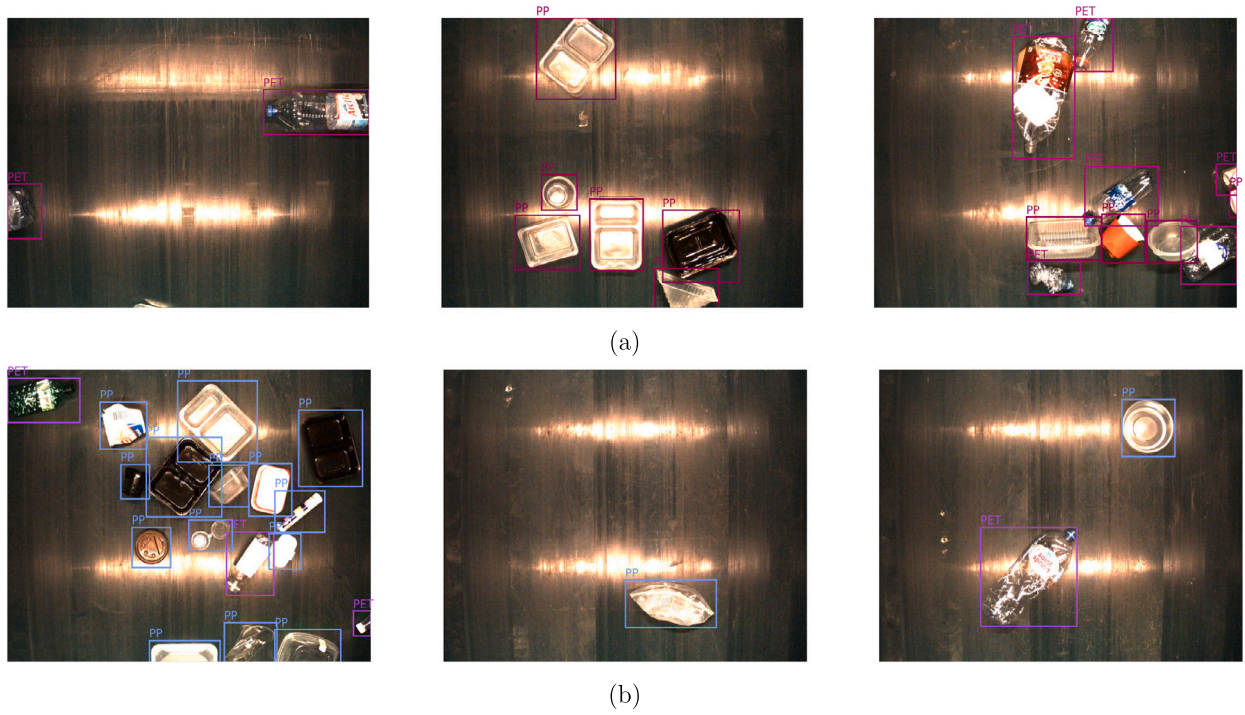


Fig. 3. Annotated samples from (a) train set and (b) test set.

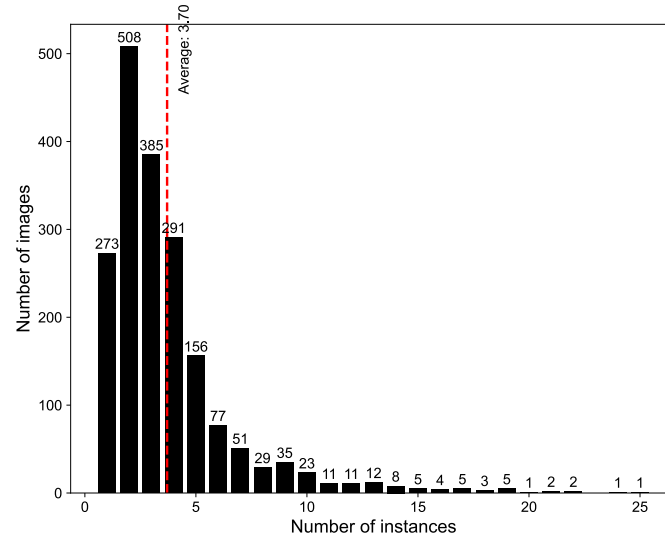


Fig. 4. Number of annotations by class.

other types of waste like glass, paper and metal is time-consuming and resource-intensive due to collecting, storing and labeling items. The selected classes of plastic waste, PET and PP, are among the most commonly used plastics in everyday products such as bottles, packaging and containers. By focusing on these classes, the aim is to create the dataset relevant to real-life application of plastic sorting. Nevertheless, further addition of plastic categories is highly recommended in order to create a robust deep learning algorithm that can tackle various scenarios.

More than 85,000 images are captured by a camera on a moving conveyor belt, containing a mix of PET and PP objects, along with the non-waste and empty images. An algorithm is implemented to remove the empty images using thresholding based on the rate of white pixels. After manually selecting the images from different frames where objects are located in various places on the conveyor, only 1600 images are used for the train/validation set and 300 images for the test set. The dataset is annotated with the bounding box and a respective plastic

category (see Fig. 3). The final dataset of 1900 images consists of a total of 7033 annotations of which 5947 belong to the train set and 1086 to the test set. Fig. 4 depicts the number of instances in images. In general, every image in average has 2 to 4 bounding boxes. The majority of the dataset images has less than 10 instances with only a small number of images containing more than 10 instances. The maximum number of instances in a single image is 25 which is an outlier and appears in only one image. The distribution of instances per image suggests that the dataset contains a diverse range of the object densities which is important for training object detection models to generalize well to different scenarios. It was the primary aim of this dataset. Fig. 5 shows bounding boxes locations and their sizes. One can see that the majority of bounding boxes is located near the center of the images and their size ranges from 100 to 300 pixels on average while the minor objects are characterized by the size less than 100 pixels and more than 400 pixels showing variety of PET and PP materials.

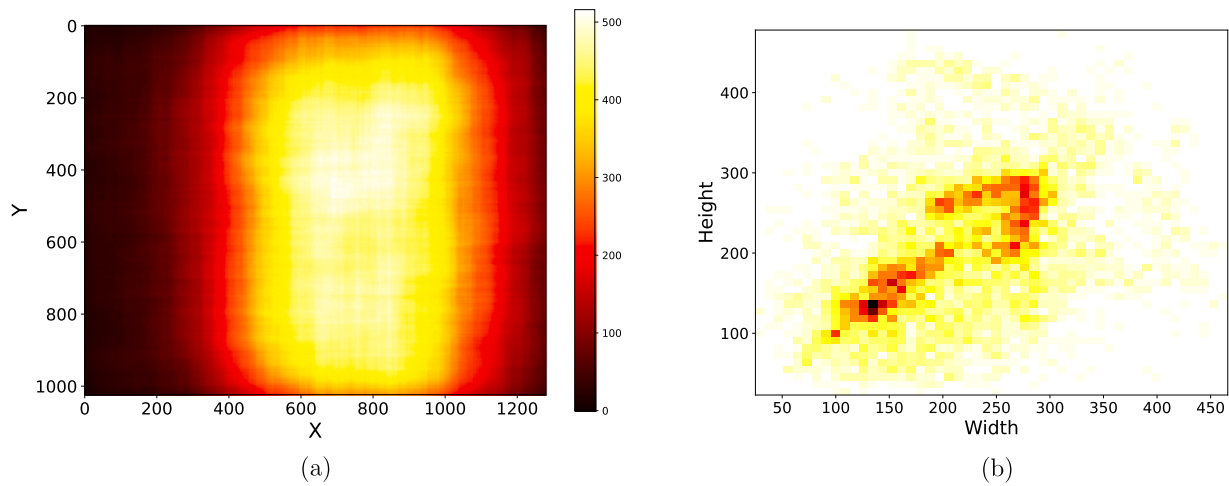


Fig. 5. (a) Frequency plot and (b) bounding box size frequency.

The annotations are saved in MS COCO (Lin et al., 2014) format while at the same time provided in YOLO (Bochkovskiy et al., 2020) format to facilitate the use of different object detection models.

3.2. Deep learning

Three object detection models Faster R-CNN and RetinaNet with different backbones and YOLOv8 were chosen for this study. All models were implemented using the PyTorch deep learning framework.

Faster R-CNN is a two-stage object detection model that uses a Region Proposal Network (RPN) to generate region proposals followed by a classification and regression network to refine the proposals and predict class labels and bounding boxes. RPN takes the images of varying sizes as input and generates the output of rectangular object proposals which in their turn have “objectness score”. The RPN creates the convolutional feature map using sliding windows of a fixed size. The feature maps are produced from the backbone models like ResNet or VGG (Ren et al., 2015). At every sliding window of each anchor box, the RPN predicts the probability of the box containing an object and the difference between the box’s position and the size compared to the actual object in the image. These predictions are used to rank the anchor boxes by their objectness score and non-maximum suppression is applied to produce a set of high-quality object proposals (Ren et al., 2015). While at the second part of the network, which takes as input proposals produced by the RPN to train Fast R-CNN network (Girshick, 2015) which is classification and regression network to predict object category and its bounding box positions. However, to get an end-to-end training process the Fast R-CNN is iteratively utilized to initialize the RPN (Ren et al., 2015).

RetinaNet (Lin et al., 2017) is a one-stage object detection model that uses a Feature Pyramid Network (FPN) and a novel focal loss function to address the problem of class imbalance in object detection that the current dataset has (1:2). Unlike Faster R-CNN, RetinaNet does not use an explicit region proposal network to generate the candidate object locations. Instead, RetinaNet uses a FPN to extract the feature maps at multiple scales from an input image. A set of anchor boxes are then placed on each scale of the feature map to predict objectness scores and bounding box offsets for each anchor. This process is performed through a shared convolutional network that predicts class probabilities and box offsets for each anchor on all the feature map scales simultaneously. This allows the model to capture features of objects at different scales, resulting in more accurate detection of objects of varying sizes. Additionally, RetinaNet introduces a loss function called focal loss, which down-weights the contribution of easy examples during training to focus more on hard examples. Overall, RetinaNet achieves competitive

accuracy with the state-of-the-art object detection models while being faster and more memory-efficient.

YOLOv8 (Jocher et al., 2023) is the state-of-the-art one-stage object detection model that can process the images and videos in real time based on the family of YOLO architectures. YOLOv8 consists of three main components: a backbone network, a detection head, and a loss function. YOLOv8 uses a new detection head called YOLO-Anchor-Free, which does not rely on predefined anchor boxes or grid cells to make predictions. Instead, it predicts four corner points for each bounding box and uses a centerness score to filter out low-quality detections. This allows YOLOv8 to handle objects of various shapes and sizes more effectively and robustly.

The choice of these models was based on their high accuracy and popularity in the object detection literature. By using three different models with different backbones, the aim is to compare their performance and determine which model is better suited for plastic object detection.

3.3. Optimization for real-time plastic waste detection

Developed deep learning algorithms must be robust in real-time inference during the waste detection process. The models are typically deployed on the embedded devices with limited computation capability. Therefore, models should be properly optimized for efficient processing of video streams without compromising accuracy. The implementation and optimization of the models were carried out on the Jetson Nano Developer Kit by Nvidia (NVIDIA, 2023a) and was intended for edge computation. The Jetson Nano, with its GPU architecture and power efficiency, offers an ideal platform for executing the deep learning tasks on the edge for this experiment.

The following pipeline was chosen to optimize the model parameters. First, PTH file from PyTorch model is rebuilt as Open Neural Network Exchange (ONNX) file of the trained model which allows more flexibility such as adding and removing some layers of neural network. ONNX is a widely adopted open-source format that enables the interoperability between different deep learning frameworks, allowing the models to be trained in one framework and deployed in another (Bai et al., 2019). Second, the models are converted into Engine file by TensorRT and deployed using DeepStream on the recorded test video.

TensorRT plays a crucial role in optimizing the performance of the deep learning models deployed on the Jetson Nano embedded device. TensorRT is a powerful tool specifically designed for accelerating and optimizing the inference on GPUs (NVIDIA, 2023b). By leveraging various optimization techniques such as layer fusion, precision calibration, and dynamic tensor memory management, TensorRT significantly enhances the efficiency and speed of inference. With the aim of achieving

the real-time waste detection on the conveyor belt, TensorRT's ability to optimize network layers and reduce memory footprint is particularly valuable (Park et al., 2021). Additionally, its support for model quantization and precision calibration ensures that the deep learning models can be efficiently deployed on the resource-constrained Jetson Nano without compromising accuracy. By integrating TensorRT into the deployment pipeline we can leverage its capabilities to achieve high-performance inference enabling real-time and efficient detection of plastic waste.

3.4. Evaluation metrics

Mean Average Precision (mAP) is a metric used for evaluation of object detection models. Average Precision (AP) incorporates the use of three metrics: precision, recall and Intersection over Union (IoU). Precision measures how accurate the positive predictions are and is defined as the ratio of true positives to the total number of positive predictions made by the model, while recall measures how well the model is able to find all the positive instances and is defined as the ratio of true positives to the total number of actual positive instances in the dataset. At the same time, IoU is a measure of the overlap between the predicted bounding box and the ground-truth bounding box for an object. IoU is often used to determine whether a prediction is a true positive or a false positive. These metrics are used to calculate the Precision-Recall (PR) curve which shows the trade-off between the precision and recall for different threshold values of IoU. AP then is the area under the PR curve using the trapezoidal rule:

$$AP = \int_0^1 p(r)dr \quad (1)$$

where r is the recall value at each point on the PR curve.

Then mAP is the average APs across all the classes:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (2)$$

where N is the number of classes which is 2 in our case.

In order to gain a deeper insight into the performance of the model, the AP is calculated at various IoU thresholds. These thresholds range from 0.50 to 0.95, as well as being calculated specifically for the range of 0.50 and 0.75. In addition to mAP metric, additional test set classification accuracy is also calculated for the best model which is the rate of correctly predicted classes over all classes:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

where TP, TN, FP and FN are the true positive, true negative, false positive and false negative, respectively.

3.5. Experimental setup

As mentioned earlier, this study focuses on three popular object detection models: Faster R-CNN, RetinaNet, and YOLOv8. These models have demonstrated robustness to class imbalance, good performance on objects of varying sizes, and reasonable computation time.

Two different backbone architectures, ResNet-50 and ResNet-101, were used for Faster R-CNN in the experiment. Two separate experiments were also conducted, one using pretrained weights for the models and the other training the models from scratch. Similarly, for RetinaNet, only one backbone architecture was used, ResNet-101, which was trained with both pretrained network weights and from scratch. The pretrained networks are trained and evaluated using the MS COCO dataset. YOLOv8 offers several model sizes that are suitable for edge computing, but for this study, the *small* and *large* models were used to train plastic waste detection. Furthermore, an additional experiment is

done with the smallest model YOLOv8 which is YOLOv8 *nano* for further model pruning.

The collected dataset discussed in Section 2.2 is employed to train and evaluate neural network models for binary object detection. For Faster R-CNN, RetinaNet, YOLOv8 models, the dataset is split into 1500, 100 and 300 for train, validation and test sets, respectively. It is worth noting that a validation set is randomly derived from the initial train set. Mild augmentations are applied that include image resizing, brightness, color contrast modifications.

Faster R-CNN is trained using a multi-task loss which is the sum of classification loss and regression loss. The classification loss measures the error between the predicted class of the object in the image, and the regression loss measures error between the predicted and actual bounding box coordinates of the objects in the image. The regression loss is calculated using smooth L1 loss. RetinaNet employs Focal loss for the classification task and Distributed Focal loss for the box regression task, while YOLOv8 uses Binary Cross Entropy (BCE) loss for classification task and DFL for the box regression task. SGD optimizer is used in Faster R-CNN and RetinaNet models, and Adam optimizer is used for YOLOv8 models. The batch size used for training is 8, and models are trained for 150 epochs. Warm-up iterations with a low learning rate are applied, followed by a change to a learning rate of 0.0001, and controlled by a WarmupMultiStepLR learning rate scheduler for Faster R-CNN and RetinaNet models, and by a Lambda learning scheduler for YOLOv8 models. The models are evaluated with mAP metrics. The experiments are conducted on a server with Intel(R) Xeon(R) Gold 6140 CPU @ 2.30 GHz CPU of 16 GB RAM and NVIDIA Tesla V100 SXM2 16 GB GPU with Linux operating system. The experiments utilize PyTorch, an open-source deep learning framework and neural networks architecture implementations from Detectron2 (Wu et al., 2019) and Ultralytics (Jocher et al., 2023). After training and evaluating, the best models according to size and metric results are employed to further quantization which includes hyperparameters optimization, TensorRT conversion and deployment in Jetson Nano.

4. Results

The learning curves are presented in Fig. 6. From these figures it can be observed that both Faster R-CNN and RetinaNet models reached a plateau in their losses at approximately 0.1 and 0.25, respectively. On the other hand, although YOLOv8 models reached the maximum number of epochs, the losses were still decreasing and reached around 0.6. This suggests that YOLOv8 models could benefit from longer training or a reduction in learning rate. However, this does not apply for YOLOv8 Nano model, since the training had stopped around the 120th epoch due to early stopping after 20 consecutive epochs for no improvements in validation metrics. It is important to note that the loss graphs of YOLOv8 only show box regression losses, while the loss graphs of Faster R-CNN and RetinaNet models represent the total loss.

For these experiments, the performance of the models is evaluated using various metrics. The primary metric used to evaluate model performance is mAP, which is calculated for different IoU thresholds ranging from 0.50 to 0.95. In addition to mAP, two other metrics are calculated: mAP₅₀, which is the mean average precision at an IoU threshold of 0.50 and mAP₇₅, which is the mean average precision at an IoU threshold of 0.75.

All metrics are calculated using a confidence threshold of 0.8, which means that only detections with a confidence score greater than or equal to 0.8 are considered. Non-maximum suppression (NMS) is also applied with an IoU threshold of 0.50 to reduce overlapping detections.

Fig. 7 shows the growth of mAP over time during the training and testing on the validation set. All models showed good mAP results, with Faster R-CNN and RetinaNet models exceeding 70% and remaining relatively stable at the end of training. In contrast, YOLOv8 models exceeded 80% and demonstrated a slight growth towards the end of training. This observation is consistent with the learning curves where

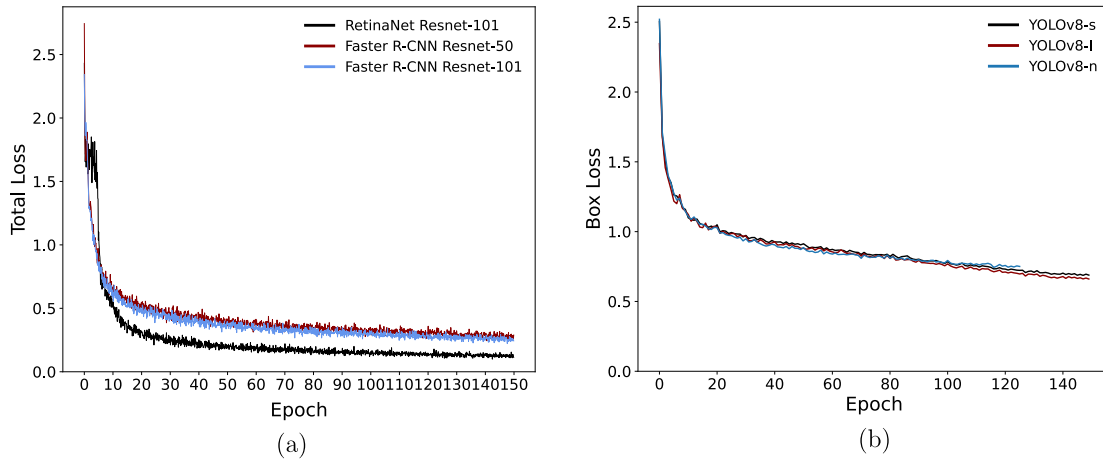


Fig. 6. Learning curves of (a) Faster R-CNN and RetinNet models (not pretrained) and (b) YOLOv8 models.

Table 2

Summary of object detection models performance on a validation set.

Model	Backbone	Pretrained	mAP	mAP ₅₀	mAP ₇₅	AP per class (PET/PP)
Faster R-CNN	Resnet50	No	73.69	96.50	88.00	73.87 / 73.52
		Yes	73.08	97.00	89.09	72.11 / 74.05
	Resnet101	No	75.54	97.05	91.06	76.18 / 74.91
		Yes	74.32	97.12	92.18	74.27 / 74.36
RetinaNet	Resnet101	No	76.04	96.60	90.84	74.81 / 77.27
		Yes	74.88	96.00	90.52	73.72 / 76.03
YOLOv8	Nano	No	81.35	93.90	92.75	83.00 / 79.70
	Small	No	82.35	95.60	93.94	82.86 / 81.84
	Large	No	82.94	95.73	94.24	83.46 / 82.41

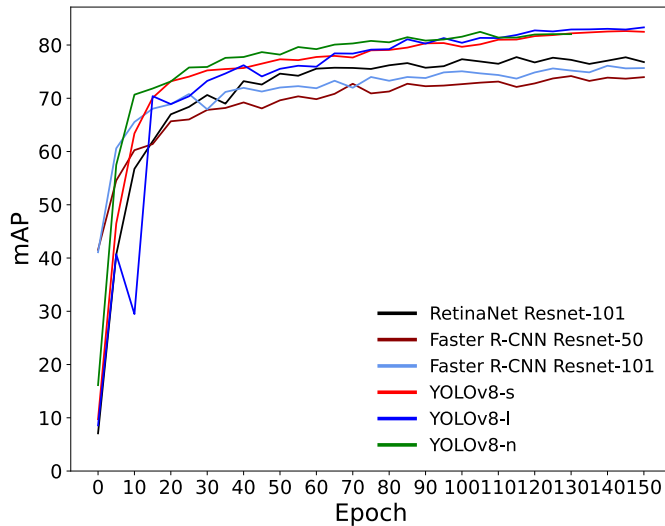


Fig. 7. mAP results of trained object detection models.

YOLOv8 models continued to make progress even after other models had plateaued.

Table 2 and Table 3 summarize the models' performance on the validation and test sets, respectively. It can be observed that the YOLOv8 Large model achieved the best results in both sets with an mAP of 82.94% and 77.74% for the validation and test sets, respectively. Notably, being the best achieving model in terms of mAP, its accuracy score is high, achieving 95.67%. The RetinaNet model with pretrained weights achieved the highest mAP₅₀ metric, outperforming other mod-

els in this specific aspect. Interestingly, upon analyzing the mAP₅₀ metric, it can be deduced that for both in validation and test sets, YOLOv8 models achieved lower results comparing to Faster R-CNN and RetinaNet models comparing to mAP and mAP₇₅. These variations can be attributed for the inherent differences in the neural network architectures. YOLOv8 focuses on faster training and inference which is well suited for real-time applications. However, this speed advantage may result in challenges when detecting small or closely packed objects, unlike the Faster R-CNN and RetinaNet models. In the context of mAP₅₀ which measures the performance at a lower IoU threshold and allows for less precise bounding box predictions, the YOLOv8 models may yield lower scores. Conversely, the mAP and mAP₇₅ metrics use higher IoU thresholds emphasizing the accurate localization and detection. Given the characteristics of YOLOv8's architecture, it tends to perform well in scenarios where objects are larger or more distinct, but may struggle with smaller or closely packed objects that require higher precision. This can explain the lower mAP₅₀ scores compared to other models that excel in precisely localizing smaller objects. Nevertheless, the differences between models in terms of mAP₅₀ are not significant, as indicated in the tables.

Before finalizing the model, learning curves suggested potential improvement if some fine-tuning were applied. The process of fine-tuning the YOLOv8 Small model involved implementing several techniques such as using pretrained model weights, applying a different optimizer, specifically AdamW, utilizing label smoothing, introducing various weight decays, implementing other learning rate schedulers and training for more numbers of epochs. Despite these efforts, the fine-tuned model did not exhibit any improvements over the originally trained model, and instead resulted in lower maximum achieved mAP scores of 82.73%, as compared to the original model's mAP score of 82.94%. Despite the lack of improvement, the exploration of various

Table 3
Summary of object detection models performance on a test set.

Model	Backbone	Pretrained	mAP	mAP ₅₀	mAP ₇₅	AP per class (PET/PP)
Faster R-CNN	Resnet50	No	66.75	90.70	80.71	66.89 / 66.61
		Yes	67.63	91.74	82.94	67.88 / 67.38
	Resnet101	No	68.74	91.71	83.33	68.63 / 68.86
		Yes	67.04	92.15	82.97	66.68 / 67.40
RetinaNet	Resnet101	No	67.48	88.17	81.86	65.99 / 68.98
		Yes	68.43	89.44	83.75	69.14 / 67.72
YOLOv8	Nano	No	75.14	86.27	85.04	76.08 / 74.21
	Small	No	77.03	88.60	87.79	77.82 / 76.24
	Large	No	77.74	88.90	88.10	78.04 / 77.44

fine-tuning techniques can provide valuable insights into optimizing model performance in the future.

4.1. Quantization results

The YOLOv8 family models showed better performance in mAP comparing to other models during evaluation. Among the YOLOv8 backbones, the Large model demonstrated the highest metrics, surpassing the Small model by 0.71% and Nano model by 2.6%. However, considering the computational efficiency and parameter size, the decision was made to optimize and implement the YOLOv8 Small and Nano models on the Jetson Nano platform. The Small model with 11.2 million parameters and the Nano model with 3.2 million parameters proved to be a suitable choice compared to the Large model which had 25.9 million parameters. The optimization ensures balance between the performance and computational efficiency for deployment on the Jetson Nano device. Furthermore, by choosing two small models it would be easy to understand the models for future deployment in real site.

In order to assess the optimization capabilities of the models, the metric used was the inference time, measured in Frames Per Second (FPS). Several experiments were conducted to optimize the models and evaluate their performance on a separate 2 minutes long test video. The baseline for comparison was the YOLOv8 small model without any model pruning which achieved an inference speed of 4.5 FPS as well as YOLOv8 nano model – 9 FPS. Additional experiments involved compressing the models further by using different floating point precision configurations including 32-bit (FP32) and 16-bit (FP16) precision. These experiments aimed to explore the trade-off between the model size and the inference speed, and to determine the most efficient configuration for real-time plastic waste detection on the Jetson Nano. To get better insights of the models' performance, Jetson Nano is set to maximum power input. Since the current version of NVIDIA Jetson Nano does not support INT8 calibration, we failed testing the performance in even more compressed model.

As a result of experiments, the best performing model in terms of FPS is YOLOv8 Nano model with 20 FPS in FP16 precision and 15 FPS in FP32 precision. Followed by YOLOv8 Small model with 10 FPS in FP16 precision and 6 FPS in FP32 precision.

Considering the operational speed of the industrial conveyor belt which moves at the speed of 3 m/s and the camera's ability to capture a 1-meter segment of the conveyor in each exposure, it can be deduced that within a span of one second a 3-meter long sequence of waste passes beneath the camera. Hence, the frame rate of 3 FPS is deemed sufficient to effectively capture and analyze the waste as it traverses along the conveyor belt. However, a higher frame rate may be desirable to ensure accurate and timely detection of plastic waste items. A faster frame rate allows for more frequent analysis of the scene, reducing the likelihood of missing or misclassifying objects. In the context of the Jetson Nano device which has limited computational power, achieving a frame rate of 20 FPS can be considered acceptable and practical.

This balance between the frame rate and device capability ensures the efficient and accurate waste detection while accommodating the constraints of the embedded system.

5. Discussion

The results and implementation of this research highlight the effectiveness of YOLOv8-based architectures in the context of plastic waste sorting. These models consistently achieved high performance, with the best model achieving mAP 77.74% and the accuracy 95.67%. The choice of YOLOv8 can be attributed to its recent development and popularity as a framework for object detection, particularly in the industrial settings. Despite the architectural and framework differences among the chosen models they all demonstrated consistent and robust results. Furthermore, the models were pruned to ensure their suitability for deployment in industrial settings, striking a balance between accuracy and inference speed. As a result, the YOLOv8 models emerged as the top performers among the models evaluated in this research. Moreover, it is important to note that higher thresholds for NMS and confidence scores were selected to optimize the models for real-life scenarios, where accurate predictions are crucial. Although there are some works that outperformed our models (Sousa et al., 2019, Wang et al., 2020) their datasets are not collected in industrial settings which substantially reduces practical feasibility for the plastic waste sorting. On a contrasting note, the results of object detection in the research conducted by Majchrowska et al. (2022) reveal a scenario where the same deep learning models achieved substantially diverse metrics across various datasets. This underscores the influence that dataset quality and representation can wield over the efficacy of waste detection capabilities. The focus of this research was to test the deep learning models on a more diverse dataset close to an industrial environment as well as to improve the processing time of the image recognition step in a plastic waste sorting pipeline.

In addition to their strong performance in terms of object detection accuracy, the developed models have also been successfully deployed on the embedded system showcasing their practical potential. With careful optimization and leveraging the capabilities of TensorRT, the models achieved the inference speed 20 FPS in FP16 precision. This real-time processing capability is crucial for the applications in SWM where timely and efficient detection of plastic waste is essential.

The successful deployment of the models on the Jetson Nano combined with their accurate predictions and real-time processing capabilities, signifies an important milestone in the research.

However, it is worth mentioning the limitations of this research. Firstly, the collected dataset only should be extended by adding more plastic categories with various level of damage and contamination to present more comprehensive representation of waste objects. Secondly, achieved frame rate could be further improved to enhance more robust performance for high speed conveyor belt. Therefore, these identified limitations provide valuable insights for future research in the field of

waste detection. By addressing these challenges further advancements can be made to enhance the applicability of deep learning algorithms in industrial solid waste management. One of the future directions could be using multi-fusion models by incorporating a dataset collected from different camera sources, like RGB camera along with NIR camera. In addition, more advanced augmentation and fine-tuning techniques could increase diversification within training dataset to achieve better performance in plastic waste detection. Finally, the waste detection task could be extended to instance segmentation task where the models need to predict in addition to bounding box and class of the object their segmentation mask. This will potentially enable better results during plastic waste sorting by the machine.

Overall, the results indicate that the developed models have the potential to be deployed in practical applications, such as solid waste management facilities to guarantee the accurate object classification for efficient sorting.

6. Conclusions

In this article, we have addressed the challenges of industrial plastic waste sorting by developing an efficient and accurate IoT solution based on deep learning algorithms and embedded intelligence. For this task, a novel dataset specifically focused on detection of plastic waste with PET and PP categories is collected. Various state-of-the-art deep learning object detection methodologies, such as Faster R-CNN, RetinaNet and YOLO, are employed to achieve high mAP metrics of 77.74%. Furthermore, the developed algorithms are thoroughly fine-tuned and optimized for later deployment on the embedded devices like Jetson Nano, resulting in a frame rate of 20 FPS. The research holds the significant milestone for future application of these findings and advancing in solid waste management offering the reduction of decreasing installation and maintenance costs, minimization of energy consumption, and enhancement of the sorting accuracy of automated waste sorting machines.

The primary research question was whether a high accuracy deep learning algorithm be developed to detect plastic wastes on a high-speed conveyor belt and facilitate replacing traditional sorting methods. The study successfully accomplished these objectives and hypotheses, as demonstrated by the achieved mAP metric and the optimized implementation on the embedded system. The study contributes to the existing knowledge by demonstrating the effectiveness of these algorithms in a real-world waste sorting scenario. The optimized implementation on embedded system provides the practical feasibility of the developed algorithms. By accurately identifying and classifying plastic waste in matter of short time, developed pipeline can contribute to the efficient management of resources and the promotion of recycling initiatives across the globe. The study fills the gap in the existing literature by focusing on the detection of specific plastic waste categories and providing insights into the implementation of these algorithms on embedded devices as well as presenting a novel plastic waste detection dataset.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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